

Note that all arguments are in radians. And here follows a demonstration of a small subset of these:

```
$ maxima -q
(%i1) cos(0);
(%o1) 1
(%i2) cos(%pi/2);
(%o2) 0
(%i3) cot(0);
The number 0 isn't in the domain of cot
-- an error. To debug this try: debugmode(true);
(%i4) tan(%pi/4);
(%o4) 1
(%i5) string(asin(1));
(%o5) %pi/2
(%i6) csch(0);
The number 0 isn't in the domain of csch
-- an error. To debug this try: debugmode(true);
(%i7) csch(1);
(%o7) csch(1)
(%i8) asinh(0);
(%o8) 0
(%i9) string(%i * sin(%pi / 3)^2 + cos(5 * %pi / 6));
(%o9) 3*%i/4-sqrt(3)/2
(%i10) quit();
```

Simplifications with special angles like $\pi/10$ and its multiples can be enabled by loading the *ntrig* package. Check the difference below before and after the package is loaded:

```
$ maxima -q
(%i1) string(sin(%pi/10));
(%o1) sin(%pi/10)
(%i2) string(cos(2*%pi/10));
(%o2) cos(%pi/5)
(%i3) string(tan(3*%pi/10));
(%o3) tan(3*%pi/10)
(%i4) load(ntrig);
(%o4) /usr/share/maxima/5.24.0/share/trigonometry/ntrig.
mac
(%i5) string(sin(%pi/10));
(%o5) (sqrt(5)-1)/4
(%i6) string(cos(2*%pi/10));
(%o6) (sqrt(5)+1)/4
(%i7) string(tan(3*%pi/10));
(%o7) sqrt(2)*(sqrt(5)+1)/((sqrt(5)-1)*sqrt(sqrt(5)+5))
(%i8) quit();
```

A very common trigonometric problem is as follows: given a tangent value, find the corresponding angle. A common challenge is that for every value, the angle could lie in two quadrants. For a positive tangent, the angle could be in the first or the third quadrant, and for a

negative value, the angle could be in the second or the fourth quadrant. So, *atan()* cannot always calculate the correct quadrant of the angle. How then, can we know what it is, exactly? Obviously, we need some extra information, say, the actual values of the perpendicular (p) and the base (b) of the tangent, rather than just the tangent value. With that, the angle location could be tabulated as follows:

Perpendicular (p)	Base (b)	Tangent (p/b)	Angle quadrant
Positive	Positive	Positive	First
Positive	Negative	Negative	Second
Negative	Negative	Negative	Third
Negative	Positive	Positive	Fourth

This functionality is captured in the *atan2()* function, which takes two arguments, 'p' and 'b', and thus does provide the angle in the correct quadrant, as per the table above. Along with this, the infinities of tangent are also taken care of. Here's a demo:

```
$ maxima -q
(%i1) atan2(0, 1); /* Zero */
(%o1) 0
(%i2) atan2(0, -1); /* Zero */
(%o2) %pi
(%i3) string(atan2(1, -1)); /* -1 */
(%o3) 3*%pi/4
(%i4) string(atan2(-1, -1)); /* 1 */
(%o4) -3*%pi/4
(%i5) string(atan2(-1, 0)); /* -Infinity */
(%o5) -%pi/2
(%i6) string(atan2(5, 0)); /* +Infinity */
(%o6) %pi/2
(%i7) quit();
```

Trigonometric identities

Maxima supports many built-in trigonometric identities and you can add your own as well. The first one that we will look at is the set dealing with integral multiples and factors of π . Let's declare a few integers and then play around with them:

```
$ maxima -q
(%i1) declare(n, integer, n, integer);
(%o1) done
(%i2) properties(m);
(%o2) [database info, kind(n, integer)]
(%i3) sin(m * %pi);
(%o3) 0
(%i4) string(cos(n * %pi));
(%o4) (-1)^n
(%i5) string(cos(m * %pi / 2)); /* No simplification */
```